# Module I — Crystal structure & X-rays

Theory notes: concepts, results and what every symbol means

## Crystal, lattice, basis

**IN ONE LINE** A crystal is one group of atoms (the **basis**) stamped onto every point of an endless regular grid (the **lattice**).

### Crystalline vs amorphous

Solids come in two kinds, and the difference is just *how far the tidiness reaches*.

- **Crystalline** — the atomic pattern repeats predictably over enormous distances. This is *long-range order*: knowing where one atom sits lets you predict one a million atoms away. Diamond, quartz, salt, copper.
  - *Single crystal* — the pattern runs unbroken through the whole piece.
  - *Polycrystalline* — many small tidy grains pointing in random directions, meeting at *grain boundaries*. Most metals you handle are like this.
- **Amorphous** — tidy over only a few atoms (*short-range order*), then it loses the plot. Glass, plastics. It happens when a liquid is cooled so fast the atoms freeze before they can line up.

Crystals form because the ordered arrangement has the **lowest energy**. Nature takes it whenever it is given time.

### Lattice, basis, structure — keep these three apart

These three words get mixed up constantly, and separating them earns marks on its own.

- **Lattice** — an imaginary set of *points*, repeating forever. No atoms in it at all. Its defining property: *every point has exactly the same view of its surroundings as every other point*.
- **Basis** — the actual atom, or group of atoms, that you put at each point.
- **Structure** — what you get when you do that:

$$\text{lattice} + \text{basis} = \text{crystal structure}$$

**WHERE**

| | |
|---|---|
| **lattice** | the endless regular array of points, all with identical surroundings (no atoms yet) |
| **basis** | the atom, or group of atoms, attached to every lattice point |
| **crystal structure** | the finished pattern of atoms in space |

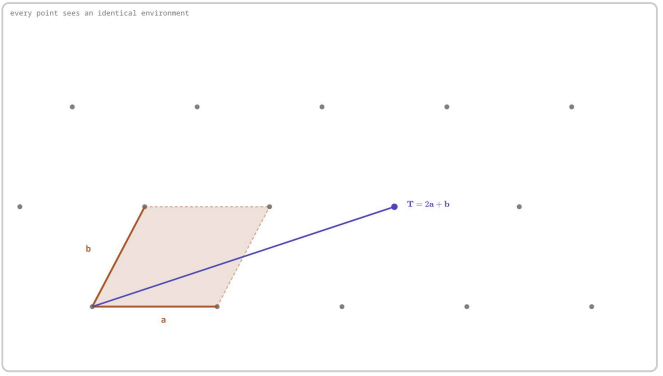
Think of patterned wallpaper. The lattice is the set of positions where the motif gets stamped; the basis is the motif; the wallpaper is the structure. Two very different wallpapers can use the same stamping positions.

You travel between lattice points using a **translation vector** $\mathbf{T}$ — a recipe of whole-number steps along three chosen directions $\mathbf{a}, \mathbf{b}, \mathbf{c}$:

$$\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c} \qquad (n_1, n_2, n_3 \text{ whole numbers})$$

**WHERE**

| | |
|---|---|
| $\mathbf{T}$ | lattice translation vector: it carries you from one lattice point to another |
| $\mathbf{a},\ \mathbf{b},\ \mathbf{c}$ | primitive vectors, the three edges of the primitive cell |
| $n_1,\ n_2,\ n_3$ | whole numbers (positive, negative or zero); each choice reaches one lattice point |




A 2-D oblique lattice with one choice of primitive vectors $\mathbf{a}, \mathbf{b}$. The shaded parallelogram is the primitive cell — one lattice point in total, since each corner is shared by four cells. The indigo arrow is a translation $\mathbf{T} = 2\mathbf{a} + \mathbf{b}$: land on it and the view around you is identical to the view from the origin. That last sentence *is* the definition of a lattice.

If whole numbers alone can reach *every* point, those vectors are called **primitive**.

### Unit cells, and counting the atoms inside one

A **unit cell** is the small box you repeat to build the whole crystal.

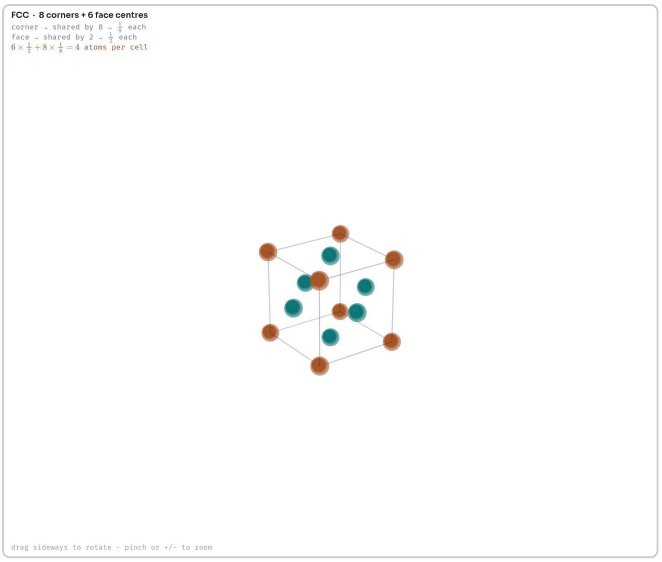
- **Primitive cell** — the smallest box possible. It works out to exactly **one** lattice point per cell, with volume $V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.
- **Conventional cell** — usually a bigger, nicer box (a cube, say) with extra points at the body centre or face centres. It holds more than one point, but we use it because it *shows the symmetry*, which a skewed primitive cell hides.

An atom on a corner or a face is *shared* with neighbouring boxes, so you only count your share. A corner is shared by 8 boxes, a face by 2, an atom fully inside belongs to you alone:

$$N = N_{\text{inside}} + \frac{N_{\text{face}}}{2} + \frac{N_{\text{corner}}}{8}$$

**WHERE**

| | |
|---|---|
| $N$ | effective number of atoms (lattice points) belonging to one cell |
| $N_{\text{inside}}$ | atoms wholly inside the cell: each counts 1 |
| $N_{\text{face}}$ | atoms on the faces: each is shared by 2 cells, so counts $\frac{1}{2}$ |
| $N_{\text{corner}}$ | atoms at the corners: each is shared by 8 cells, so counts $\frac{1}{8}$ |




**Spin it.** Why the fractions are there. Step 2 draws in the eight cells that meet at one corner — the atom is one of theirs as much as one of yours, so you may count only $\frac{1}{8}$ of it. Step 3 does the same for a face atom with its single neighbour. Rotate while the neighbour cells are showing; the sharing is much easier to believe when you can see it from behind.

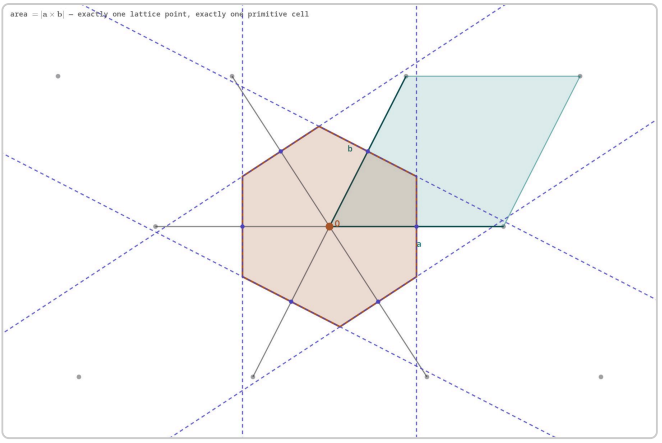
**LEARN THIS BY HEART**

**Simple cubic $N = \frac{8}{8} = 1$ · Body-centred cubic (BCC) $N = 1 + \frac{8}{8} = 2$ · Face-centred cubic (FCC) $N = \frac{6}{2} + \frac{8}{8} = 4$**

One more cell is worth a sentence. The **Wigner–Seitz cell**: stand on a lattice point, draw a line to each neighbour, and cut every line in half with a perpendicular wall. The region you are left standing in is the cell — primitive *and* fully symmetric. Remember that recipe: the identical construction done in reciprocal space later gives the Brillouin zone.

### Drawing a Wigner–Seitz cell — do it once by hand

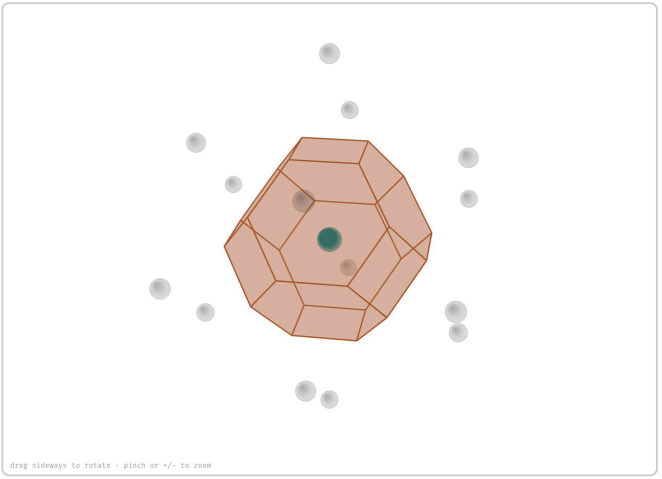
Draw it yourself in 2-D once and you own it for good, because the 3-D cell and every Brillouin zone later is the same four moves: *join, halve, bisect, keep the middle*. Step through the construction, then switch the lattice underneath and watch the same four moves give a different shape.




The construction one step at a time. A square lattice gives back a square and a hexagonal one a hexagon, but the oblique lattice gives an irregular hexagon — still primitive, still one point, still the full symmetry. Watch the faint bisectors: they belong to further-out neighbours and never reach the cell, which is exactly how you know you have drawn enough of them.

**WHERE MARKS GET LOST**

Two ways this goes wrong in an exam. **Bisecting the wrong thing:** the bisector is perpendicular to the *line joining the two points* and passes through its midpoint — not perpendicular to a cell edge. **Stopping too early or too late:** include a neighbour and its bisector may simply sit outside and do nothing (harmless), but leave out a near one and the cell comes out too big. Safe habit: bisect every neighbour out to about $1.5\times$ the nearest distance, then keep the innermost region.




**Spin it.** The Wigner–Seitz cell of a BCC lattice — the region closer to the central point than to any neighbour. The app builds it by actually cutting with all those perpendicular walls, so the shape you see (8 hexagon faces + 6 square faces) is a result, not a drawing.

## Symmetry & the 5-fold ban

**IN ONE LINE** A crystal can only have 1, 2, 3, 4 or 6-fold rotation symmetry — anything else cannot fill space without leaving gaps.

### The four symmetry operations

A **symmetry operation** is any move that leaves the crystal looking exactly as it did. There are four kinds:

1. **Translation** — slide by $\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}$.
2. **Rotation** — turn by $\theta = 2\pi/n$ about an axis. We call it an $n$-*fold axis*: $n = 4$ means a quarter turn restores it.
3. **Reflection** ($m$) — mirror it through a plane.
4. **Inversion** ($\bar{1}$) — push every point through a centre to the opposite side, $\mathbf{r} \to -\mathbf{r}$.

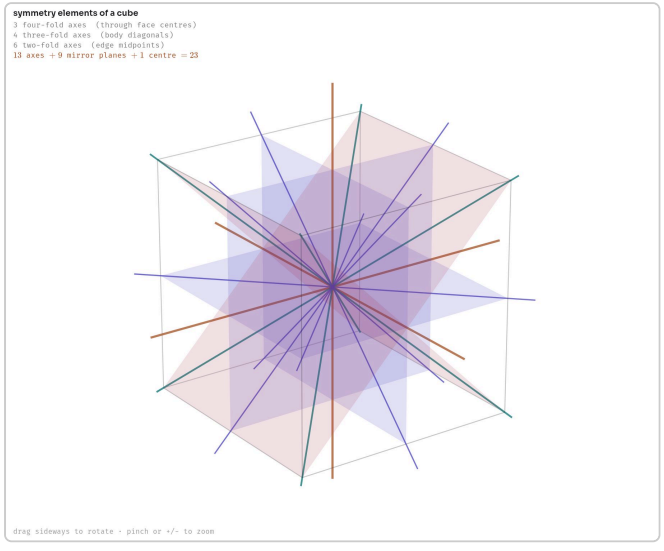
Collect the operations that leave one point fixed (everything except translation) and you have a **point group**. Add translations too and you get a **space group**.

**LEARN THIS BY HEART**

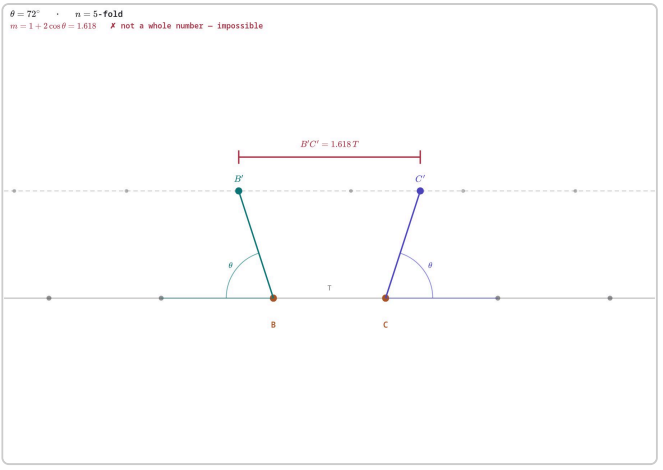
2-D: **10** point groups, **17** space groups. 3-D: **32** point groups, **230** space groups. Four cheap marks.

### Why 5-fold symmetry is impossible

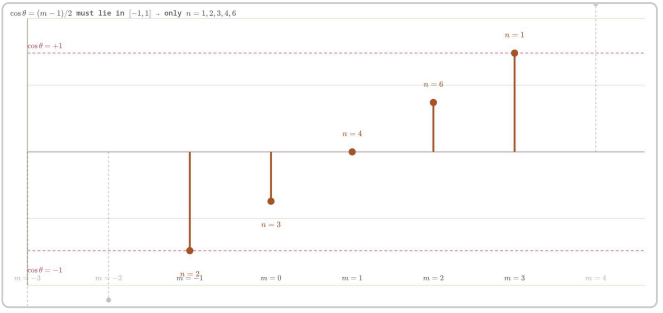
You can tile a floor with triangles, squares or hexagons and leave no gaps — but never with regular pentagons. Crystals face the same restriction, and there is a short proof.




**Spin it.** Before the ban, see what a cube *does* have. Step through the three families of rotation axis and rotate the cube to sight straight down each one — looking down a four-fold axis you see a square, down a body diagonal a triangle, and the reason each turn works becomes obvious. Then the mirrors appear and the count comes to 23. Nowhere in that list is a five-fold axis, and the next figure proves it never can be.




The proof is a picture before it is algebra. Turn the row about B one way and about C the other, and the two new points must land on a lattice row — so the gap between them has to be a whole number of steps. Try each angle: $60^\circ, 90^\circ, 120^\circ, 180^\circ$ all give whole numbers, $72^\circ$ gives $1.618$ and fails. That number is the golden ratio, which is exactly why quasicrystals — ordered but *not* periodic — are the loophole.




The proof in one picture. Each stem is $\cos\theta = (m-1)/2$ for a whole number $m$. Only the five landing inside the $\pm 1$ band are possible, and they give exactly $n = 1, 2, 3, 4, 6$. A 5-fold axis would need a value outside the band, so it cannot exist in a repeating crystal.

## Bravais lattices

**IN ONE LINE** Once only the legal symmetries are allowed, exactly 7 box shapes and 14 distinct lattices survive in 3-D. This is a table to memorise.

### The table

A box is described by three edge lengths $a, b, c$ and the three angles between them $\alpha, \beta, \gamma$. Sorting boxes by symmetry gives **7 crystal systems**. Allowing extra points at the body centre ($I$), all face centres ($F$) or one pair of faces ($C$) gives **14 Bravais lattices**.

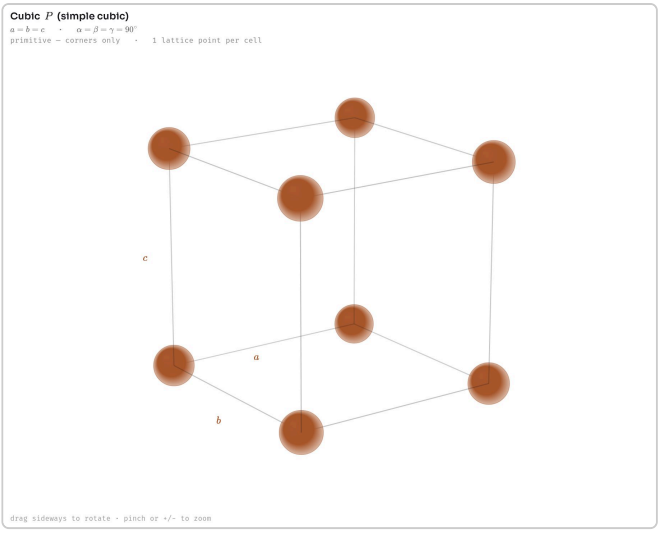
| System | Lengths | Angles | Lattic |
|---|---|---|---|
| Cubic | $a = b = c$ | all $90°$ | $P, I,$ (3) |
| Tetragonal | $a = b \neq c$ | all $90°$ | $P, I$ ( |
| Orthorhombic | $a \neq b \neq c$ | all $90°$ | $P, I,$ (4) |
| Rhombohedral | $a = b = c$ | equal, not $90°$ | $R$ (1) |
| Hexagonal | $a = b \neq c$ | $90, 90, 120°$ | $P$ (1) |
| Monoclinic | $a \neq b \neq c$ | two $90°$, one not | $P, C$ |
| Triclinic | $a \neq b \neq c$ | all different | $P$ (1) |

$$3 + 2 + 4 + 1 + 1 + 2 + 1 = 14$$

In 2-D the answer is **4 systems**, **5 lattices** — oblique, rectangular *primitive*, rectangular *centred*, square, hexagonal. The rectangular system is the one that supports two.

### All 14, in the round

The table is only half the memory. What actually sticks is the *shape*: which box got squashed, which got sheared, and where the extra points sit. Walk the whole list below — pick a system, then a centring — and say the row of the table out loud as each cell appears.




**Spin it.** Every one of the 14. Choose a **system** on the top row and a **centring** underneath — the greyed-out letters are the centrings that give nothing new for that system, which is exactly why the count stops at 14. Copper balls are the corner points, teal balls the points centring adds, and the teal shading marks *which* faces carry them. Watch the box itself change: cubic → tetragonal stretches $c$; → orthorhombic makes all three edges differ; → monoclinic shears one angle; → triclinic shears all three. The hexagonal cell is drawn inside the faint prism that three of them build.

Two things to take from the spin. First, a corner point is **shared by 8 cells** and a face point by 2, so $P$ owns $8 \times \frac{1}{8} = 1$ point, $I$ owns $1 + 1 = 2$, $C$ owns $1 + 2 \times \frac{1}{2} = 2$ and $F$ owns $1 + 6 \times \frac{1}{2} = 4$ — the counts printed on each cell. Second, the seven systems are one chain of *lowering* symmetry: start from the cube and stretch, then unequalise, then shear, and you arrive at triclinic, which keeps nothing but the repetition itself.

**WHERE MARKS GET LOST**

"Why is there no face-centred *tetragonal* lattice?" is a favourite. Answer: because it is **not new** — a face-centred tetragonal cell can be redrawn as a smaller *body-centred* tetragonal cell (rotate the axes $45°$ and shrink by $\sqrt{2}$). A centring only counts as a separate Bravais lattice if it cannot be re-described as an existing one. That test is what trims the list to 14.

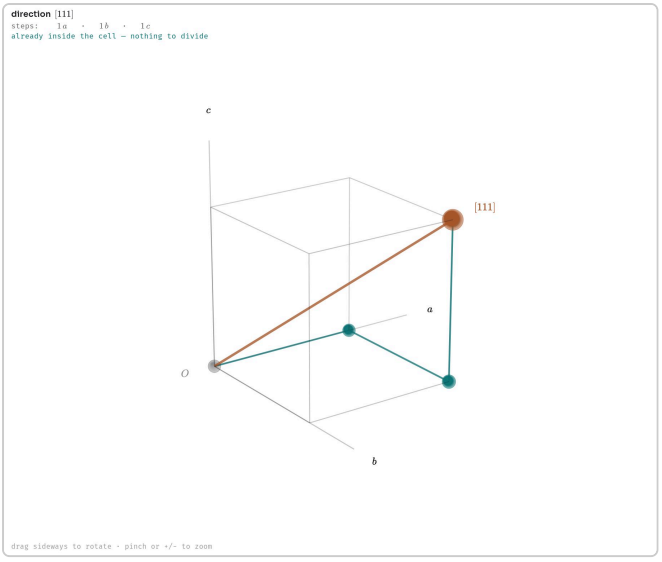
## Directions & Miller indices

**IN ONE LINE** Directions get square brackets $[uvw]$. Planes get round brackets $(hkl)$ and are labelled by the **reciprocals** of where they cut the axes.

### Naming a direction: [uvw]

Draw the arrow from the origin, read its components in units of $a, b, c$, and scale to the smallest whole numbers. Square brackets.

- $[uvw]$ — one specific direction. A minus sign is written as a bar on top: $[\bar{1}00]$ is the $-x$ direction.
- $\langle uvw \rangle$ — the whole *family* that symmetry makes equivalent. In a cube, $\langle 100 \rangle$ means all six cube-edge directions.




**Spin it.** Indices are **steps, not coordinates**. Walk $\boldsymbol{u}$ along $\boldsymbol{a}$, then $\boldsymbol{v}$ along $\boldsymbol{b}$, then $\boldsymbol{w}$ along $\boldsymbol{c}$ (teal), and the arrow from O to where you land is $[\boldsymbol{uvw}]$. Drag to spin it — a direction only becomes obvious once you have seen the arrow from two or three angles. $[211]$ shows the halving trick; $[\bar{1}10]$ steps backwards along $\boldsymbol{a}$, into the cell behind.

In a **cubic** crystal the angle between two directions is just the ordinary angle between two vectors:

$$\cos\theta = \frac{h_1h_2 + k_1k_2 + l_1l_2}{\sqrt{h_1^2 + k_1^2 + l_1^2}\,\sqrt{h_2^2 + k_2^2 + l_2^2}}$$

| WHERE | |
|---|---|
| $\theta$ | angle between the two planes (the angle between their normals) |
| $h_1k_1l_1,\ h_2k_2l_2$ | Miller indices of the first and second plane (cubic crystal) |

### Naming a plane: Miller indices (hkl)

Four steps, every single time:

1. Find where the plane cuts the three axes, measured in units of $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$. Call them $p, q, r$.
2. Take the **reciprocals**: $1/p,\ 1/q,\ 1/r$.
3. Multiply through to clear fractions, giving smallest whole numbers.
4. Write them in round brackets, no commas: $(hkl)$.

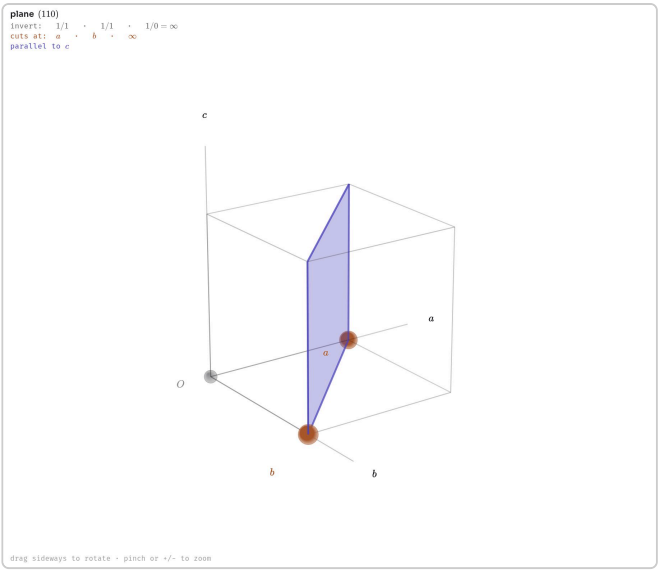
A plane *parallel* to an axis never cuts it, so that intercept is $\infty$ — and $1/\infty = 0$. That is the whole reason for step 2: reciprocals turn an unusable infinity into a perfectly good zero.

*Worked example.* Intercepts $\boldsymbol{a},\ 2\boldsymbol{b},\ \infty \to p, q, r = 1, 2, \infty$ → reciprocals $1, \frac{1}{2}, 0$ → multiply by 2 → $(210)$.

Brackets matter: $(hkl)$ is one set of parallel planes; $\{hkl\}$ is the symmetry-equivalent family.

### Drawing $(hkl)$ — the same recipe, run backwards

The question comes the other way round just as often: *"sketch the* $(110)$ *plane in a cubic cell"*. Naming runs intercepts → reciprocals; drawing runs reciprocals → intercepts, then you join them. Four moves: *invert, mark, join, shade*.




**Spin it.** Pick a plane and walk the five steps, then **spin it** — a plane drawn flat is easy to misread, and rotating shows immediately whether it really misses the axis it is supposed to be parallel to. $(100)$ comes out as a whole cube face because two intercepts are $\infty$; $(111)$ slices a corner off symmetrically; $(210)$ cuts $\boldsymbol{a}$ at $\boldsymbol{a}/2$ and sits closer to the origin; $(\bar{1}10)$ shows what a bar does — the cut moves to $-\boldsymbol{a}$, in the cell behind.

Three habits that keep the marks:

- **Mark the origin first.** A plane that passes *through* the origin has an intercept of $0$, and $1/0$ is not an index — shift the origin one cell along, or draw the next parallel plane of the same family instead. They have the same name.
- **A zero index means "never cuts".** Do not look for an intercept: run the plane parallel to that axis, right through the cell.
- **A bar means the negative side.** Step into the cell behind the origin rather than trying to squeeze the plane into the first one.

LEARN THIS BY HEART

**In a cubic crystal, the direction $[hkl]$ is perpendicular to the plane $(hkl)$.** This one fact turns most plane questions into simple vector arithmetic.

And a direction $[uvw]$ **lies inside** the plane $(hkl)$ exactly when

$$hu + kv + lw = 0$$

WHERE

| | |
|---|---|
| $(hkl)$ | Miller indices of the plane |
| $[uvw]$ | indices of the direction |
| $hu + kv + lw = 0$ | the test: if it is zero the direction lies in the plane, or parallel to it |

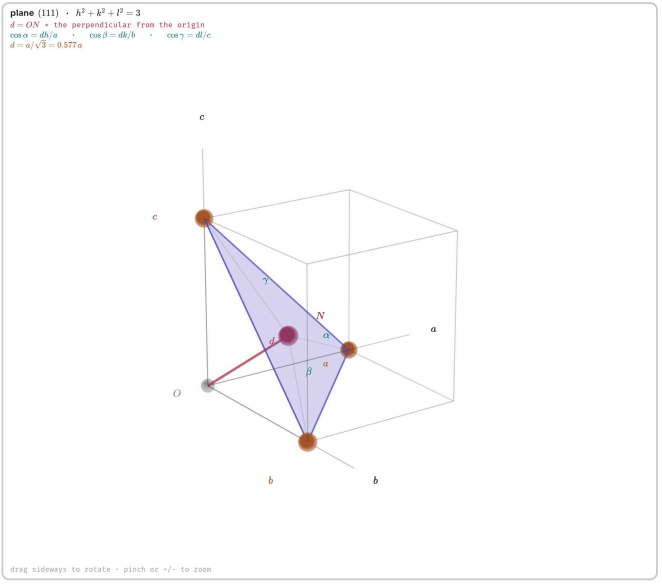
(because "lies in the plane" means "perpendicular to the plane's normal"). No drawing required.

## Interplanar spacing

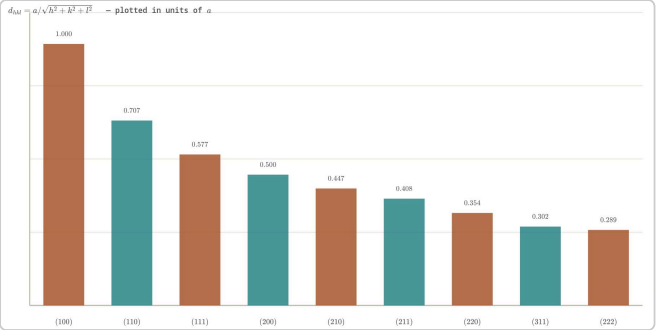
IN ONE LINE Planes with bigger Miller indices sit closer together: $d = a/\sqrt{h^2 + k^2 + l^2}$ for a cube.

### The formula, and where it comes from

A set of parallel planes $(hkl)$ has a fixed perpendicular gap between neighbours, written $d_{hkl}$. Every X-ray question needs it, so learn the cubic version cold.




**Spin it.** The whole derivation is one perpendicular. Drop $ON$ from the origin onto the nearest plane — that length *is* $d$. Rotate until you are sighting along $ON$: the plane goes edge-on and you can see that $d$ really is the gap to the next plane. Each intercept then makes a right-angled triangle, so each direction cosine is $d$ over an intercept, and the identity finishes it.




The spacings of the first few families in a cubic crystal, in units of $a$. Two things to take away: bigger indices always mean *closer* planes, and the sequence of $\sqrt{h^2+k^2+l^2}$ values — $1, \sqrt{2}, \sqrt{3}, 2, \sqrt{5}\ldots$ — is exactly what a powder pattern measures. Which entries go *missing* is how a diffraction pattern tells BCC from FCC.

Notice $d_{222} = \frac{1}{2} d_{111}$: multiplying every index by $n$ divides the spacing by $n$. That fact gets used the moment we reach Bragg's law.

## Packing & structures

**IN ONE LINE** Treat atoms as hard spheres. Where they touch fixes $a$ in terms of $r$ — and that gives you the coordination number and the fraction of space filled.

### The two numbers that describe a packing

- **Coordination number ($CN$)** — how many nearest neighbours actually touch a given atom.
- **Packing fraction ($f$)** — what fraction of the box is really filled with sphere:

$$f = \frac{(\text{atoms per cell}) \times \frac{4}{3}\pi r^3}{a^3}$$

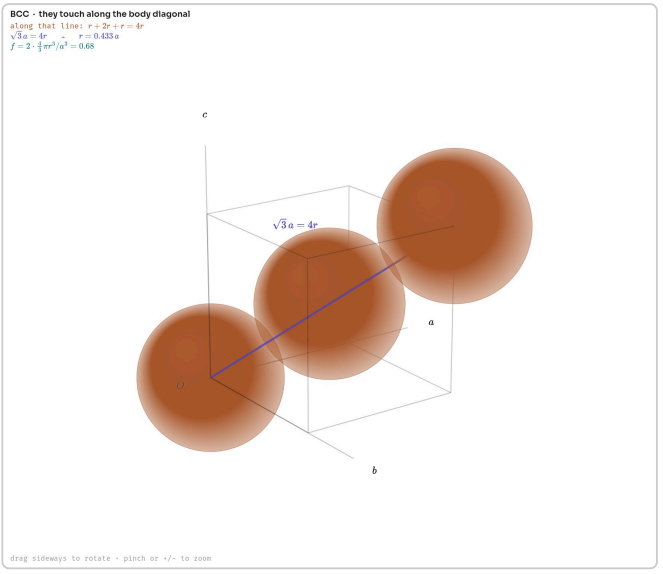
**WHERE**

| | |
|---|---|
| $f$ | packing fraction: the share of the cell volume filled by atoms |
| $r$ | atomic radius (atoms treated as touching hard spheres) |
| $\frac{4}{3}\pi r^3$ | volume of one atom |
| $a^3$ | volume of the cubic cell |

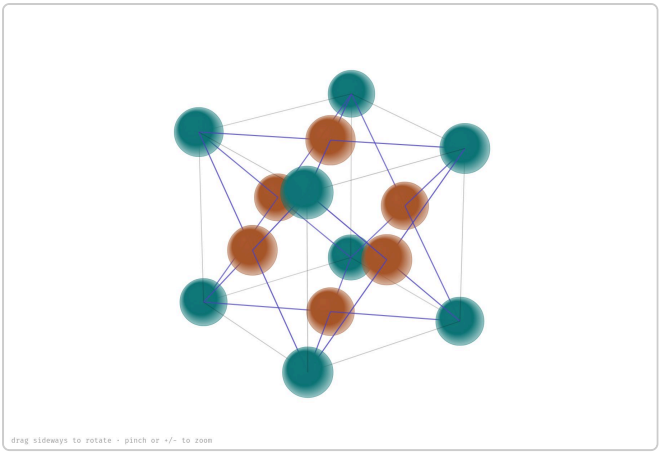
Everything follows from one question: *along which line do the spheres touch?* Answer that and you have $a$ in terms of $r$; the rest is arithmetic.

**LEARN THIS BY HEART**

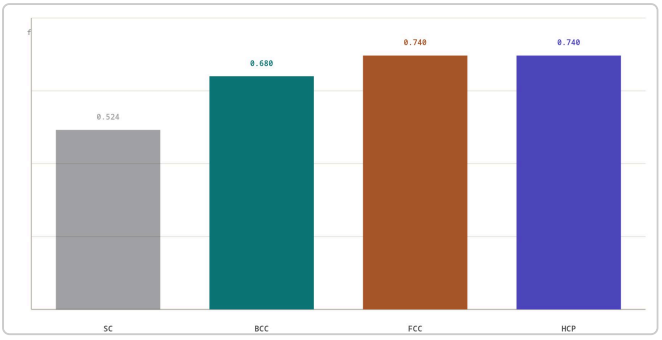
| Structure | Atoms/cell | Touch along | $a$ vs $r$ |
|---|---|---|---|
| Simple cubic | 1 | cube edge | $a = 2r$ |
| BCC | 2 | body diagonal | $\sqrt{3}\,a =$ |
| FCC | 4 | face diagonal | $\sqrt{2}\,a =$ |
| HCP | 6 | basal edge | $a = 2r$ |




**Spin it.** Where $a = 2r$, $\sqrt{3}a = 4r$ and $\sqrt{2}a = 4r$ actually come from. Pick a cell, follow the touching line, blow the atoms up, then count the radii along it. **Rotate until you look straight down that line** — the spheres line up and the count stops being a memory test. Every packing-fraction and density numerical starts from this one equation.



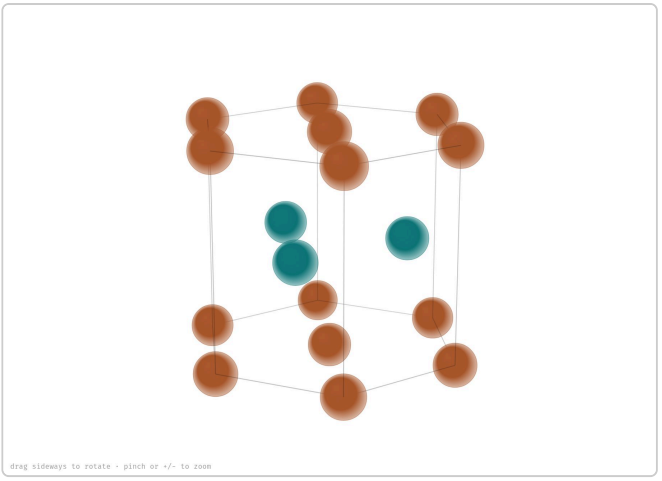

**Spin it.** The three cubic cells. Switch between SC, BCC and FCC, then press ⬤ **fill** to blow the atoms up to their real touching size — that is the moment the packing fraction becomes obvious. The indigo bonds show the touching direction: cube edge for SC, body diagonal for BCC, face diagonal for FCC.



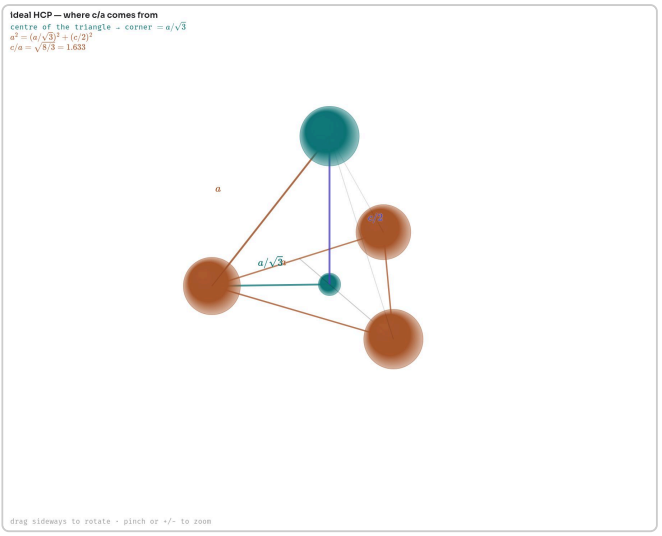

The four packing fractions side by side. FCC and HCP tie at 0.74 — the tightest possible packing of equal spheres. BCC manages 0.68, and simple cubic only 0.52.

### Close packing and the c/a ratio

FCC and HCP are both the *tightest possible* stacking of equal spheres — both give $f = 0.74$ and $CN = 12$. They differ only in the **stacking order** of the close-packed layers: HCP repeats $ABABAB\ldots$, FCC repeats $ABCABC\ldots$.



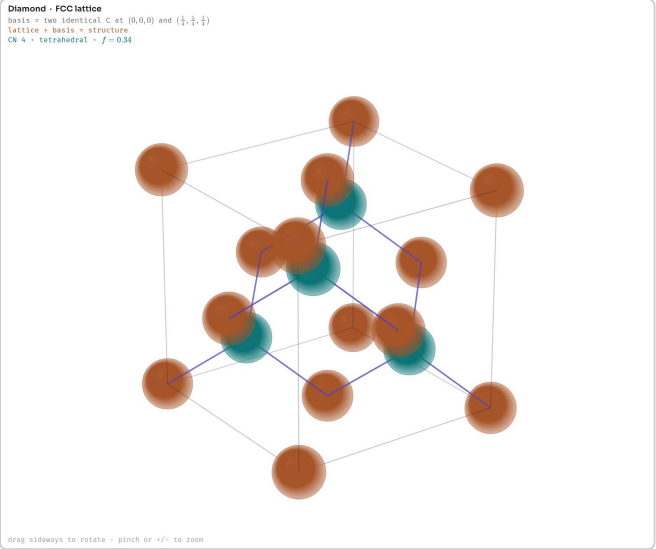

**Spin it.** Hexagonal close packing. Copper atoms are the two $A$ layers (top and bottom); the three teal atoms are the $B$ layer resting in the hollows between them. Rotate to look straight down the axis and you will see the $B$ atoms drop neatly into the gaps.




**Spin it.** One tetrahedron, and the ratio falls out. Three touching spheres make an equilateral triangle of side $a$; the hollow at its centre is $a/\sqrt{3}$ from each corner; the sphere resting there is $a$ away from each of them and $c/2$ higher. **Spin it until the right-angled triangle goes edge-on** — that is the triangle Pythagoras is applied to.

## Diamond & NaCl

**IN ONE LINE** All three are an FCC lattice with a two-atom basis — only the basis changes.



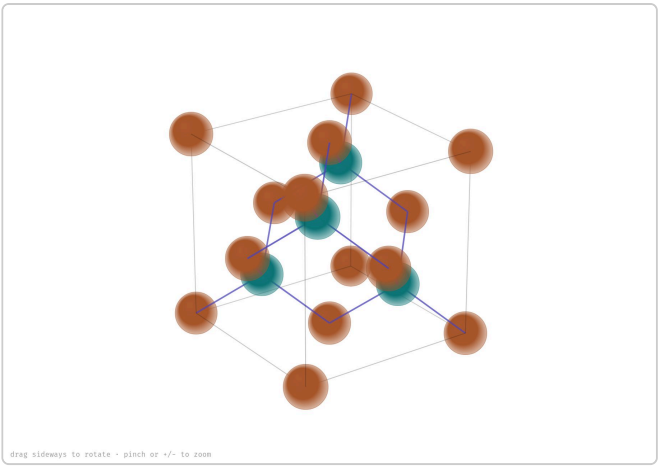

**Spin it.** The sentence "lattice + basis = structure" as a construction. The same four FCC points every time; only the basis vector changes — $(\frac{1}{4},\frac{1}{4},\frac{1}{4})$ along the body diagonal gives tetrahedral diamond and zinc blende, $(\frac{1}{2},0,0)$ along an edge gives octahedral rock salt. Spin each one and count the neighbours around a single atom: 4 for diamond, 6 for NaCl.

### Diamond and zinc blende

- **Diamond** — FCC lattice; basis = two *identical* atoms at $(0,0,0)$ and $(\frac{1}{4},\frac{1}{4},\frac{1}{4})$. So $4\times 2 = 8$ atoms per cube. Each atom has just 4 neighbours arranged tetrahedrally, at distance $\frac{\sqrt{3}}{4}a$. Packing fraction only $0.34$. Materials: C, Si, Ge, grey tin.
- **Zinc blende (ZnS)** — identical geometry, but the two basis atoms are *different* elements. Coordination $4{:}4$, 4 formula units per cube.

**WHERE MARKS GET LOST**

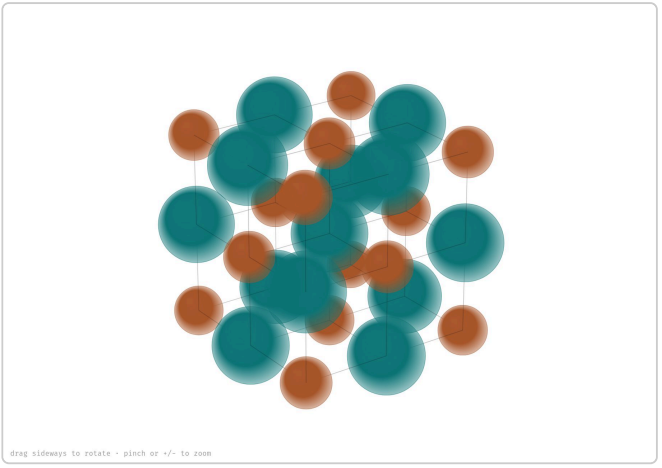
"Why does diamond fill only 34% of space when FCC fills 74%?" Because its bonding is **directional**. Covalent bonds insist on the tetrahedral $CN = 4$ arrangement, and tetrahedra cannot pack tightly. Close packing ($CN = 12$) is what you get when bonding has no preferred direction, as in metals. Structure follows bonding.




**Spin it.** The diamond cell. Copper spheres are the FCC positions, teal spheres the four basis atoms tucked into tetrahedral gaps, and every indigo bond is exactly $\frac{\sqrt{3}}{4}a$ long. Rotate until a teal atom faces you and you will see its four bonds fanning out tetrahedrally — the $CN = 4$ that keeps diamond so empty.

### Rock salt (NaCl)

- FCC lattice; basis = one $Na^+$ at $(0,0,0)$ and one $Cl^-$ at $(\frac{1}{2},0,0)$ — displaced along a cube **edge**, not the diagonal.
- 4 $Na^+$ + 4 $Cl^-$ per cube (4 formula units). Coordination $6{:}6$, octahedral.
- The ions touch along the edge, so $a = 2(r_+ + r_-)$ — the line you need for every NaCl numerical.




**Spin it.** Rock salt. Large teal spheres are $Cl^-$, small copper ones $Na^+$, alternating along every cube edge so each ion is surrounded by six of the other kind. Rotate along an edge to see the strict alternation that gives $a = 2(r_+ + r_-)$.

## X-rays & reciprocal space

**IN ONE LINE** X-rays reflect off crystal planes only at special angles ( $2d\sin\theta = n\lambda$). Describing that in "reciprocal space" turns the condition into a clean geometric picture.

## Bragg's law

To resolve something of size $d$ you need a wave of comparable wavelength. Atomic spacings are about $1$ Å, which is the X-ray range — that is the entire reason crystallography uses X-rays and not visible light.

Picture the planes $(hkl)$, spaced $d$ apart, as a stack of half-silvered mirrors. A ray reflecting off the second plane travels further than one off the first. Geometry makes that extra distance $2d\sin\theta$, and you get a bright reflection only when it equals a whole number of wavelengths:

**LEARN THIS BY HEART**

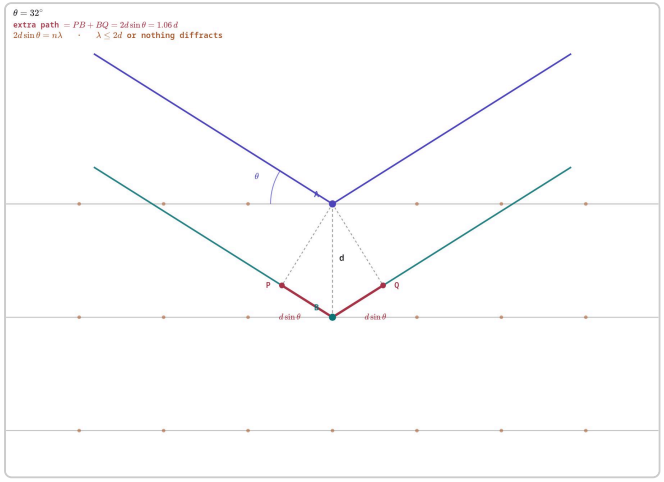
$$2d\sin\theta = n\lambda$$

**WHERE**

- $d$ spacing between the reflecting planes
- $\theta$ glancing angle: between the X-ray beam and the planes (not to the normal)
- $n$ order of the reflection: 1, 2, 3, ...
- $\lambda$ wavelength of the X-rays

$\theta$ is measured **from the plane**, not from the normal — that is the classic slip. $n$ is the order of the reflection.

Because $d_{nh\,nk\,nl} = d_{hkl}/n$, an $n$-th order reflection from $(hkl)$ can always be relabelled a first-order reflection from $(nh\,nk\,nl)$ — which is why tables list **200** and **220** reflections.




Built one line at a time. The two perpendiculars from $A$ are the whole trick: everything outside them is common to both rays, so the extra journey is just $PB + BQ$, and each of those is $d\sin\theta$. Change $\theta$ with the chips and watch the extra path grow — at small angles it is a fraction of $d$, which is why low-angle reflections need long wavelengths.

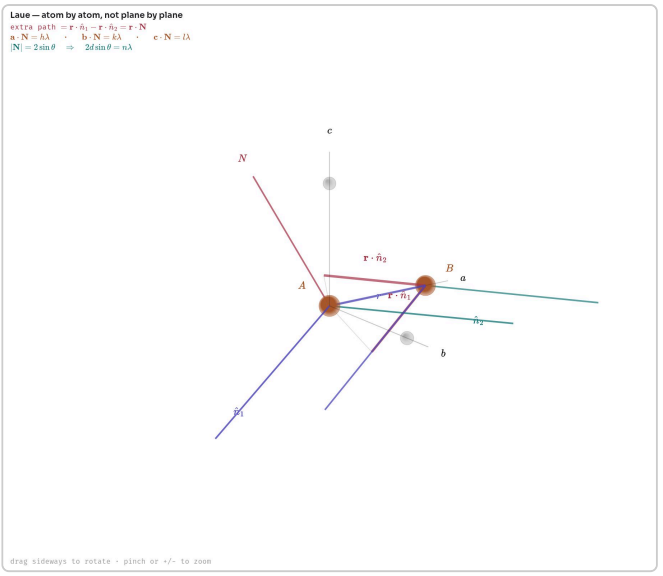
## Laue's version — and why it is the same law

Laue looked atom by atom instead of plane by plane. For two scatterers separated by $\mathbf{r}$, with $\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2$ the incoming and outgoing directions, the extra path is $\mathbf{r}\cdot(\hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2) = \mathbf{r}\cdot\mathbf{N}$. Demanding a whole number of wavelengths along *each* crystal axis gives **Laue's three equations**:

$$\mathbf{a}\cdot\mathbf{N} = nh\lambda, \qquad \mathbf{b}\cdot\mathbf{N} = nk\lambda, \qquad \mathbf{c}\cdot\mathbf{N} = nl\lambda$$

**WHERE**

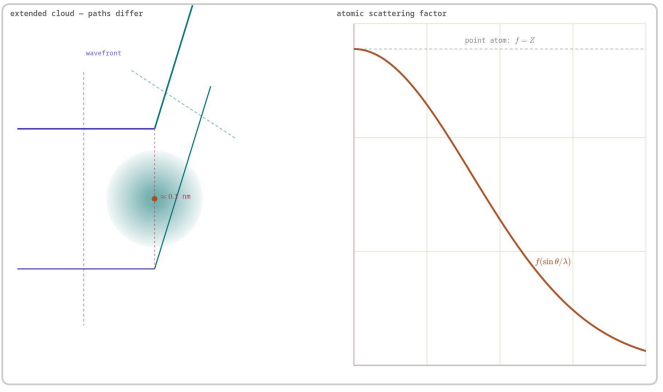
- $\mathbf{a},\ \mathbf{b},\ \mathbf{c}$ primitive vectors of the lattice
- $\mathbf{N}$ $\hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2$: difference of the unit vectors along the incident and scattered beams
- $h,\ k,\ l$ whole numbers; they turn out to be the Miller indices of the reflecting planes
- $n$ order of the reflection
- $\lambda$ X-ray wavelength




**Spin it.** Laue's argument has nothing to do with planes — just two atoms and one path difference. Step through it, then use the chips to put the second atom on each axis in turn: that is where the *three* equations come from, one per axis, all of which must hold at once. Rotate the picture at step 4 and watch the two crimson segments: whichever way you look at it, the extra journey is $\mathbf{r}\cdot\hat{\mathbf{n}}_1 - \mathbf{r}\cdot\hat{\mathbf{n}}_2$.

Rewriting these with direction cosines and $|\mathbf{N}| = 2\sin\theta$ turns them into $d_{hkl} = \dfrac{n\lambda}{2\sin\theta}$, i.e. $2d\sin\theta = n\lambda$. **Bragg and Laue are the same physics in different clothing** — a standard short-answer question.

One refinement worth a line: real atoms are fuzzy clouds of electrons, not points. Waves scattered from different parts of the same cloud partly cancel, and cancel more as the angle grows. This is the **atomic scattering factor**, and it is why high-angle reflections come out systematically weaker.




Why $f$ falls off. A point would scatter everything in step at every angle; a cloud $\sim 0.1$ nm across does not, and the mismatch grows with the scattering angle. Hence the curve on the right — and hence the systematic fading of high-angle reflections, which is a favourite one-mark question.

### The three X-ray diffraction methods

Bragg's law $2d\sin\theta = \lambda$ (first order, every plane real or imagined) has three quantities. An experiment fixes one or two and lets the crystal pick the rest. Since $\sin\theta < 1$, we need $\lambda < 2d$; with $d \sim 3$ Å that means $\lambda \lesssim 6$ Å.

| Method | Sample | X-rays | Fixed / varie |
|---|---|---|---|
| **Laue** | single crystal, held still | white (0.2–2 Å) | $\theta$ fixed, $\lambda$ varies |
| **Rotating crystal** | single crystal on a spindle | monochromatic | $\lambda$ fixed, $\theta$ varies |
| **Powder (Debye–Scherrer)** | fine powder, ~ $10^{12}$ crystallites/ $\mathrm{mm}^3$ | monochromatic | $\lambda$ fixed; $\theta$ and $d$ both vary |

**Laue:** a flat-plate camera with films for transmitted and back-reflected spots. Each set of planes picks out the wavelength that satisfies Bragg's law, so you get spots; a crystal with a 4-fold axis along the beam gives a 4-fold pattern. It cannot give the full structure, because several wavelengths and orders can land on one spot.

**Rotating crystal:** one axis (say $c$) along the rotation axis, film on a cylinder round it. Planes $(hk0)$ give the central layer line, $(hk1)$ the line above, and so on; the layer-line spacing measures the repeat distance along the axis.

**Powder:** for each $d$ the reflected rays from all the randomly oriented crystallites lie on a **cone of semi-vertical angle $2\theta$**; the cones cut the film strip in arcs. Measure the arc spacing $S$ on a camera of radius $R$: $4\theta = S/R$, then $d$ from Bragg's law.

> **WHERE MARKS GET LOST**
>
> In the powder method the cone half-angle is $2\theta$ (not $\theta$), and the distance between the two arcs of one cone corresponds to $4\theta$.

### The reciprocal lattice

Here is the idea in plain terms. A family of parallel planes is fully described by two things: which way it faces, and how far apart the planes are. So represent the whole family by a single *arrow* — pointing along the plane normal, with length inversely proportional to the spacing:

$$\mathbf{G}_{hkl} = \frac{2\pi}{d_{hkl}}\,\hat{\mathbf{n}}$$

> **WHERE**
>
> $\mathbf{G}_{hkl}$ — reciprocal lattice vector belonging to the planes $(hkl)$
> $d_{hkl}$ — spacing of those planes
> $\hat{\mathbf{n}}$ — unit vector perpendicular to the planes

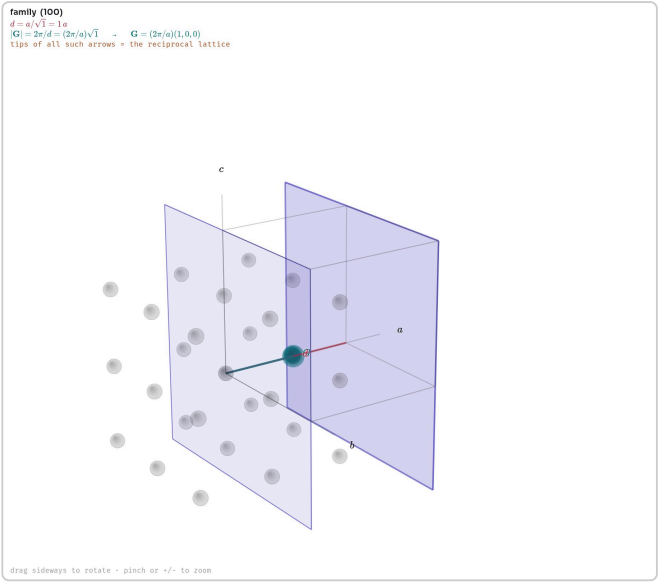
Collect the arrows for every possible family and their tips form a new lattice — the **reciprocal lattice**. It lives in "$k$-space", where distances have units of 1/length, not in ordinary space. Closely-spaced planes → long arrow → point far from the origin.

Its building vectors come from the real ones:

$$\mathbf{a}^* = 2\pi\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})}, \qquad \text{and cyclically for } \mathbf{b}^*, \mathbf{c}^*$$

> **WHERE**
>
> $\mathbf{a}^*,\ \mathbf{b}^*,\ \mathbf{c}^*$ — primitive vectors of the reciprocal lattice
> $\mathbf{a},\ \mathbf{b},\ \mathbf{c}$ — primitive vectors of the real lattice
> $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$ — volume $V$ of the real primitive cell
> $\times$ — vector (cross) product; "cyclically" means $\mathbf{b}^*$ uses $\mathbf{c}\times\mathbf{a}$ and $\mathbf{c}^*$ uses $\mathbf{a}\times\mathbf{b}$

built so that $\mathbf{a}^*\cdot\mathbf{a} = 2\pi$ while $\mathbf{a}^*\cdot\mathbf{b} = \mathbf{a}^*\cdot\mathbf{c} = 0$. The reciprocal cell has volume $(2\pi)^3/V$.




**Spin it.** The definition, made concrete. One family of planes goes in; one arrow — and therefore one point — comes out. Switch families with the chips: $(200)$ has half the spacing of $(100)$, so its arrow is twice as long and its point sits twice as far out. Do that for every family and the tips fill the faint lattice you can see at step 4. Rotate it: the reciprocal lattice of a cubic crystal is itself cubic.

LEARN THIS BY HEART

**Reciprocals of the cubic lattices — a near-guaranteed question:**

- simple cubic → simple cubic (spacing $2\pi/a$)
- BCC → FCC
- FCC → BCC

### Properties of the reciprocal lattice

1. Each reciprocal lattice point stands for one set of parallel planes of the direct lattice.
2. Its distance from the origin is inversely proportional to the spacing of those planes ($|\mathbf{G}| = 2\pi/d$).
3. The volume of the reciprocal unit cell is inversely proportional to that of the direct cell ($(2\pi)^3/V$).
4. The reciprocal unit cell need not be a parallelepiped: one usually takes its Wigner–Seitz cell, the Brillouin zone.
5. The direct lattice is the reciprocal of its own reciprocal. SC is self-reciprocal; BCC and FCC are reciprocal to each other.

## Brillouin zones

Now combine the two ideas. Diffraction happens when the wave's change of direction matches a reciprocal lattice arrow, $\mathbf{k}' = \mathbf{k} + \mathbf{G}$. The scattering is elastic so $|\mathbf{k}'| = |\mathbf{k}|$; squaring gives

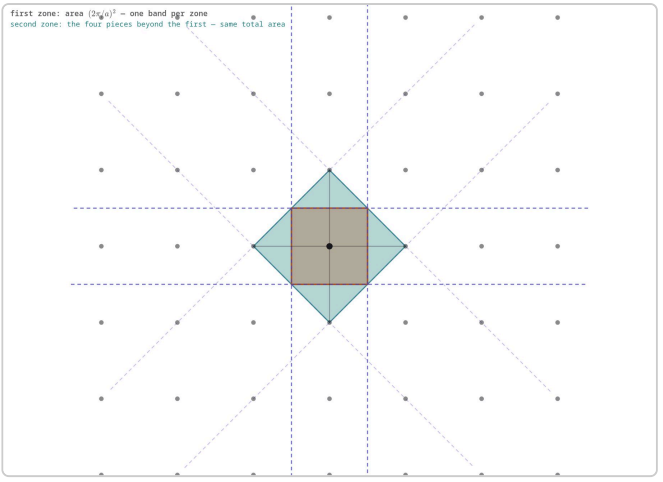
$$2\mathbf{k}\cdot\mathbf{G} + G^2 = 0 \qquad \Longleftrightarrow \qquad \mathbf{k}\cdot\frac{\mathbf{G}}{2} = \left|\frac{\mathbf{G}}{2}\right|^2$$

WHERE

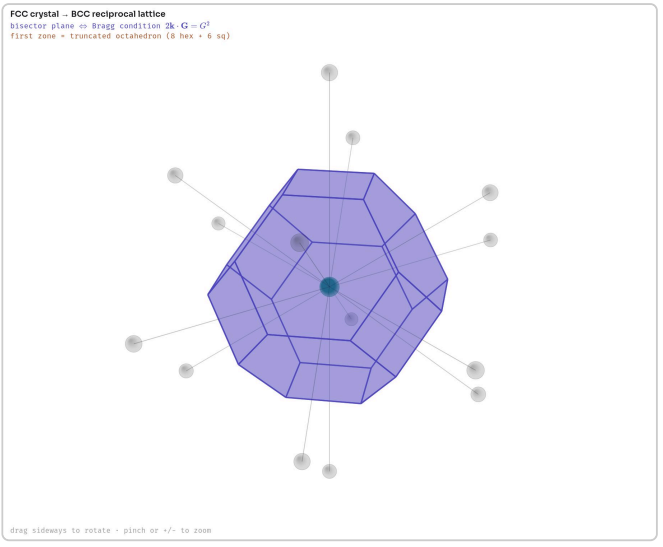
| | |
|---|---|
| $\mathbf{k}$ | wave vector of the incident wave, $\lvert\mathbf{k}\rvert = 2\pi/\lambda$ |
| $\mathbf{G}$ | a reciprocal lattice vector |
| $G$ | its length $\lvert\mathbf{G}\rvert$ |
| **meaning** | $\mathbf{k}$ ends on the plane that bisects $\mathbf{G}$ at right angles: that is the Bragg condition |

Read geometrically that says: **a wave diffracts exactly when the tip of $\mathbf{k}$ lands on the perpendicular bisector plane of a reciprocal lattice vector.**

Those bisector planes act as walls. The region around the origin enclosed by the nearest walls is the **first Brillouin zone** — which is simply the Wigner–Seitz cell from earlier, built in reciprocal space instead of real space.




The same four moves as the Wigner–Seitz cell, now in reciprocal space. Step through it: points, then the vectors G, then their perpendicular bisectors, then the region no bisector cuts off — the first zone. Step 5 adds the diagonal bisectors and the second zone appears as the teal ring. Both have area $(2\pi/a)^2$, which is why every zone holds the same number of states — one band per zone.




**Spin it.** The same construction in 3-D. An **FCC crystal** has a BCC reciprocal lattice, and its first zone is a **truncated octahedron** (8 hexagon faces + 6 squares). Switch to **BCC crystal** and you get a **rhombic dodecahedron** (12 diamond-shaped faces). Rotate and count the faces — those two names are exam answers.